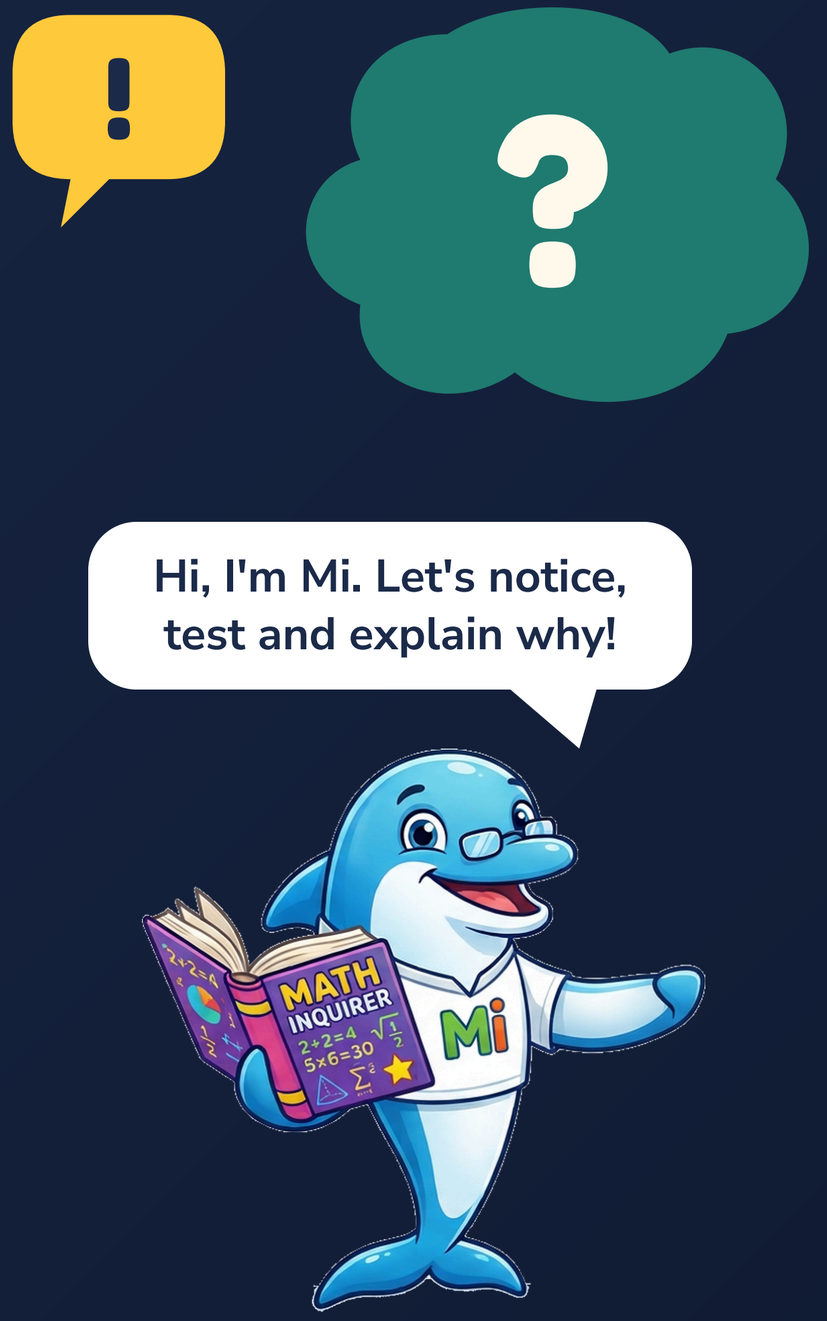
LESSON 4 • TERM 2 • LEVEL 1

# Odd + Odd?

Odd + Odd: Always, Sometimes, Never?

TODAY WE WILL

Test a claim like a detective.



2–8 MIN

# Warm-up: make pairs

**Show it on your fingers**



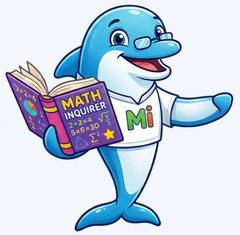
8 counters



5 counters

Is one counter left over? Flat hand for none, 1 finger for one.

**8** | 5 | **10** | 7




Fingers ready? Let us go!

# Count in twos

2–8 MIN

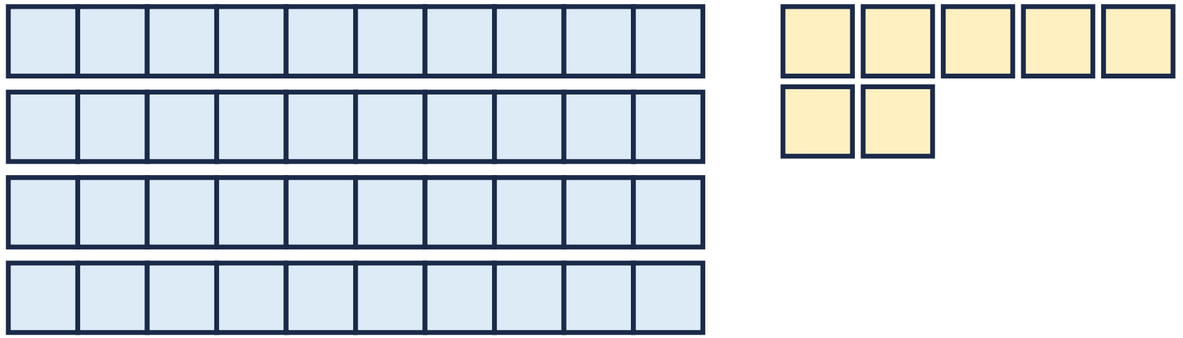
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |

Count in twos: 2, 4, 6, 8 ...

**after 10** **after 16** **before 8**

# The last digit

2–8 MIN



**47 is 4 tens and 7 ones**

The last digit of 47 is 7. It tells the ones.

**38** **63** **90**

# Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

## PUZZLE 1: anyone can try

Mei has 7 counters. She puts them in pairs.

How many counters are left over?

You may draw to help.

## PUZZLE 2: think

Ben has 5 counters. Mei has 7 counters.

They put all their counters together and make pairs.

Is one counter left over? Circle Yes or No. Then write or draw how you know.

# Explore: pair up

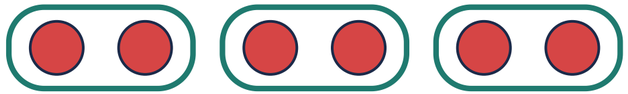
12–26 MIN

1. Take 6 counters. Put them in pairs.
2. Is one left over?
3. Take 7 counters. Put them in pairs.
4. Is one left over?
5. Try 8, 9 and 10.

# Explore: pair up

12–26 MIN

1. Take 6 counters. Put them in pairs.
2. Is one left over?
3. Take 7 counters. Put them in pairs.
4. Is one left over?
5. Try 8, 9 and 10.



6: even



7: odd

# Sort the number cards

12–26 MIN

Lay out your cards 1 to 20. Make two piles: even on the left, odd on the right.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |

**Look at the last digits. What do you notice?**

# Be a detective: test the sums

12–26 MIN

3 + 5

3 + 4

4 + 6

Build each sum with counters first.

# Be a detective: test the sums

12–26 MIN

**3 + 5** **3 + 4** **4 + 6**

| Sum | Make pairs. Odd or even? |
|---|---|
| 3 + 5 | 8: even |
| 3 + 4 | 7: odd |
| 4 + 6 | 10: even |

**Build each sum with counters first.**

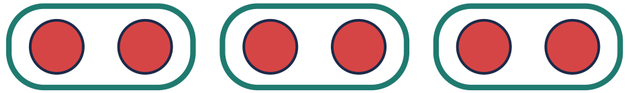
# What did we notice?

12–26 MIN

- Did odd + odd give an odd number?
- What happened with odd + even? With even + even?
- What could we test next?

# Odd and even

26–39 MIN



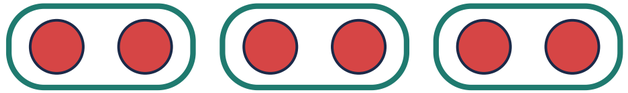
6: even



7: odd

# Odd and even

26–39 MIN



6: even



7: odd

| Last digit | The number is |
| --- | --- |
| **0, 2, 4, 6, 8** | even |
| **1, 3, 5, 7, 9** | odd |

# Test a claim: odd + even

26–39 MIN

CLAIM

An odd number plus an even number is always odd.

| Test | Total |
| --- | --- |
| 3 + 4 | 7 |
| 5 + 8 | 13 |
| 7 + 2 | 9 |

# One test can show a claim is false

26–39 MIN

CLAIM

An odd number plus an odd number is odd.

**Always, sometimes or never?**

# One test can show a claim is false

26–39 MIN

CLAIM

An odd number plus an odd number is odd.

**3 + 5 = 8. 8 is even. The claim is not always true.**

Always, sometimes or never?

# Test a claim: even + even

26–39 MIN

CLAIM

An even number plus an even number is always even.

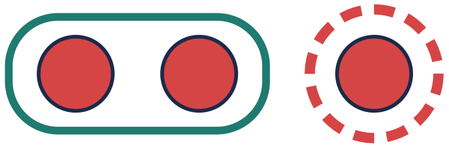
4 + 6

8 + 2

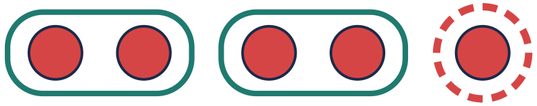
10 + 10

# Why? Draw the pairs

26–39 MIN



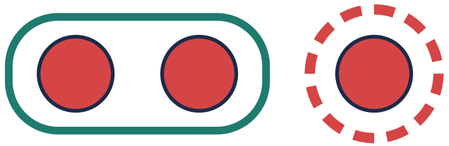
3: 1 left over



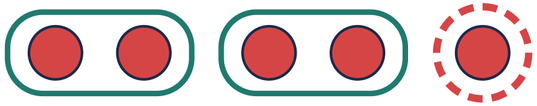
5: 1 left over

# Why? Draw the pairs

26–39 MIN



3: 1 left over



5: 1 left over

**The 2 left-over counters make a pair. So 3 + 5 has none left over: even.**

# Your turn: odd or even?

26–39 MIN

Write the total, then odd or even.

| | | | |
|---|---|---|---|
| 4 + 9 | 6 + 8 | 7 + 5 | 12 + 3 |

# Check: odd or even?

26–39 MIN

| Sum | Total | Odd or even? |
| --- | --- | --- |
| 4 + 9 | 13 | odd |
| 6 + 8 | 14 | even |
| 7 + 5 | 12 | even |
| 12 + 3 | 15 | odd |

39–43 MIN

# Clap or tap

I say a number. Even: clap twice. Odd: tap your knees once.

Say odd or even out loud.

Stand up. Sip some water.

# Workbook page 1: Start here

43–71 MIN

Questions 1 to 3. Take your time.

**S1** Look at the pairs. Fill in the missing numbers.



**S2** Pair up counters or draw dots in pairs. Circle Odd or Even for each number.

**S3** Count in 2s. Colour each number you say. Then fill in the missing number.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |

# Check page 1

43–71 MIN

| | |
|---|---|
| S1 | 6 counters: 0 left over; 7 counters: 1 left over |
| S2 | 4 even; 9 odd; 10 even; 13 odd |
| S3 | The last digits are 2, 4, 6, 8 and 0 |

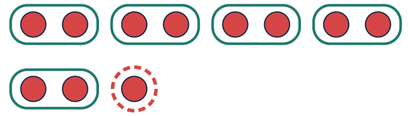
# Workbook page 2: Start here

43–71 MIN

Questions 4 to 6. Take your time.

**S4** Which number is even? Write its number in the bracket.

**S5** Look at the 11 counters in pairs. Fill in the blank. Then circle Odd or Even.



**S6** Read each statement. Decide if it is True or False.

# Check page 2

43–71 MIN

| | |
|---|---|
| S4 | (3) 14 |
| S5 | 1 counter has no partner; 11 is odd |
| S6 | True, False, True, False |

43–71 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | Test claims with sums |
| **Go further (page 4)** | Draw why it works |
| **Investigation (page 5)** | Numbers next to each other |
| **Check** | Finish and check your own work |

**Everyone may choose. Not finished? Pages 3 to 5 are for home too.**

# Talk and explain

71–77 MIN

RAVI TESTS 3 + 5

He gets 8.

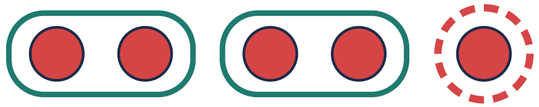
He says: odd + odd is always even.

Is one test enough? What could we do to be more sure?

I think … because …

# Back to the sneak-peek

71–77 MIN



5



7

**No! The 2 left-over counters make a pair.**

# Exit ticket

77–83 MIN

On your own. No looking back.

**E1** Is 17 odd or even? Circle Odd or Even.

17 is

**E2** Which number is even? Write its number in the bracket.

**E3** Can you make 14 by adding two odd numbers? Circle Yes or No.

14

# Check the exit ticket

77–83 MIN

| | |
|---|---|
| E1 | Odd |
| E2 | (3) 18 |
| E3 | Yes |

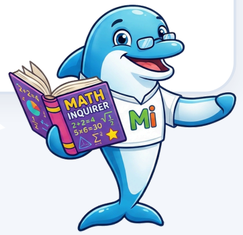
# Look-back

83–86 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ telling odd from even ☐ testing the sums<br>☐ always, sometimes or never ☐ drawing the pairs (Go further)<br>☐ something else |
| **What helped me** | ☐ counters in pairs ☐ drawing it ☐ trying again ☐ I asked for help<br>☐ I was not stuck |
| **Next time I will try** | ☐ drawing it ☐ using objects ☐ checking my answer<br>☐ asking a question |
| | ☐ I am still stuck (my grown-up will photograph this page for my tutor)<br>☐ I would like to pass |

**Take your time. You can pass.**



86–90 MIN

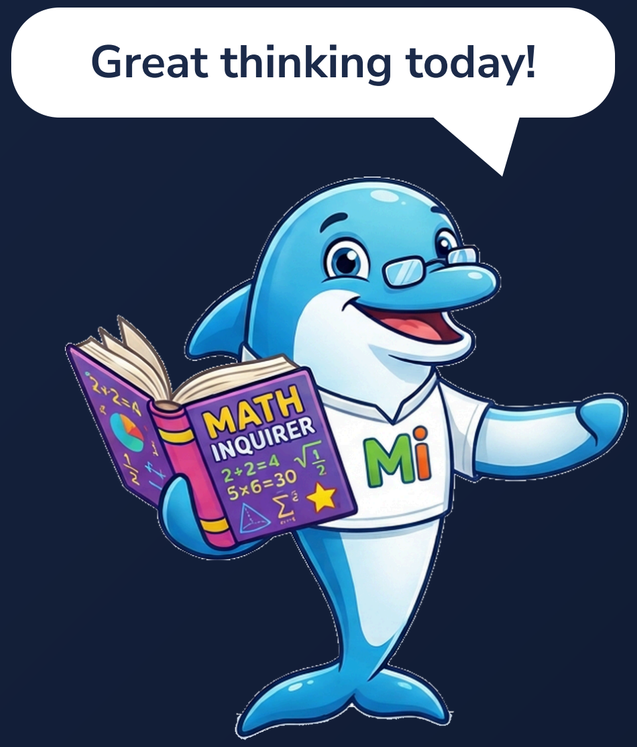
# Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

## Equal Groups

Have ready: your Term 2 Lesson 5 workbook, 4 plates or bowls, about 20 small things of one kind, pencil and paper.



# Go further: what if?

OPTIONAL

Ali says he can make 15 by adding two odd numbers. Can he?

| Try | Total |
| --- | --- |
| 7 + 7 | 14 |
| 5 + 9 | 14 |
| 7 + 9 | 16 |

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | Even + even is even. Test it three times. |
| **Make a puzzle** | Write three sums. Ask your grown-up: odd or even, before adding? |
| **Teach it back** | Show with pairs why 7 + 9 is even |

**Optional. Everyone may join.**